

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CELL

LECTURE-3

ZOOLOGY

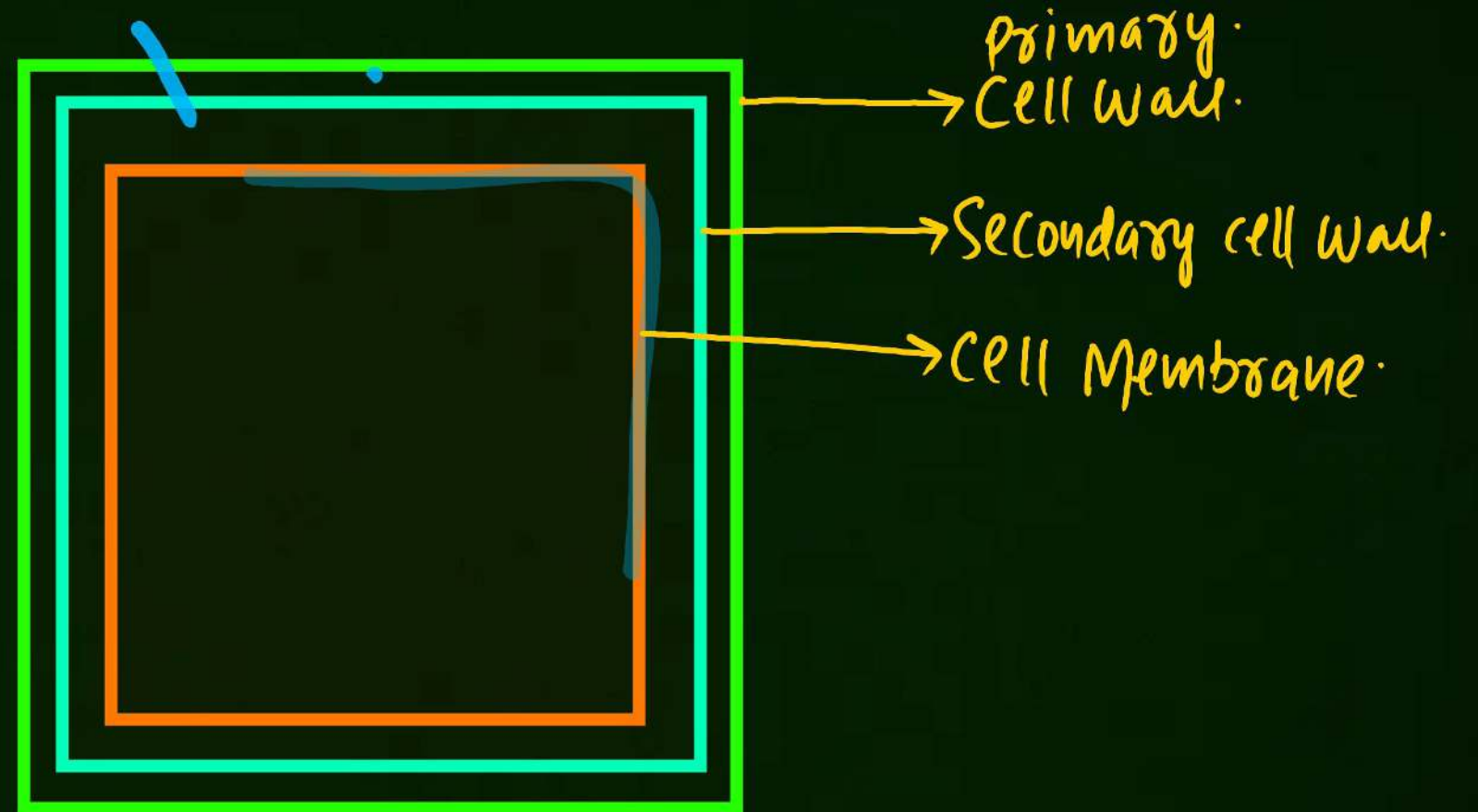
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Topics To Be Covered

- 1 Cell Wall
- 2 Endomembrane System
- 3 Previous Neet Questions
- 4

Cell Wall



Cell Wall

Structure:

- The cell wall is a rigid, non-living layer that surrounds the plasma membrane of a cell.

Composition:

- ① ○ **Plants** (Made up of cellulose, hemicellulose, pectins and proteins)
- ② ○ **Algae** (made of cellulose, galactans, mannans and minerals like calcium carbonate).
- ③ ○ **Fungi.** (made up of Chitin).

The cell wall of a young plant cell, the primary wall is capable of growth, which gradually diminishes as the cell matures and the secondary wall is formed on the inner (towards membrane) side of the cell.

Functions of Cell Wall

Functions:

- The cell wall has many functions, including:
 - **Protection:** It protects the cell from physical damage, infection, and osmotic bursting.
 - **Structure:** It gives the cell shape and structure.
 - **Regulation:** It regulates the flow of information between cells and the expansion of cells and provides barrier to undesirable macromolecules.
 - **Transport:** It allows elements to pass through the cell from a permeable membrane.

Question: NEET-1991

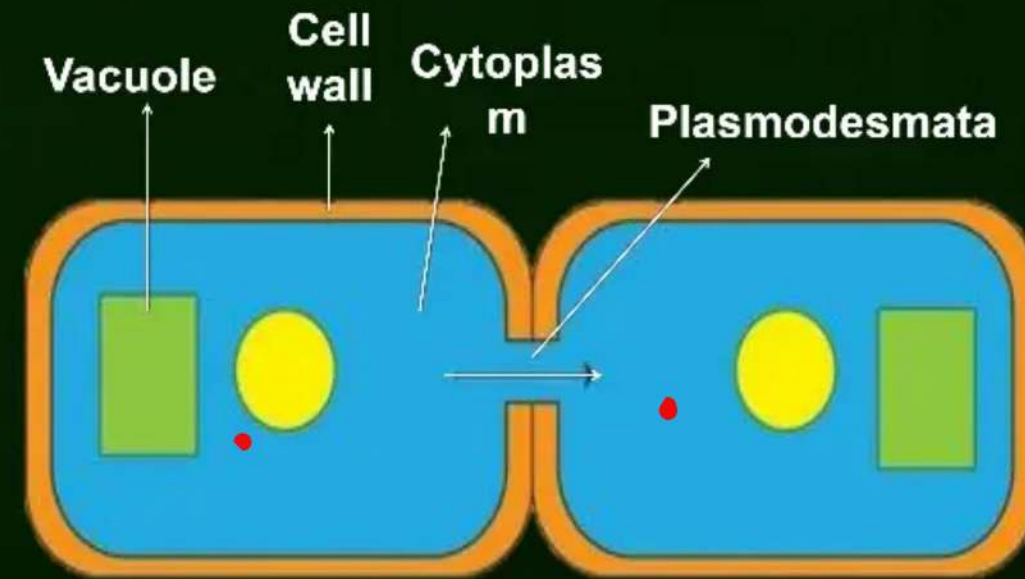
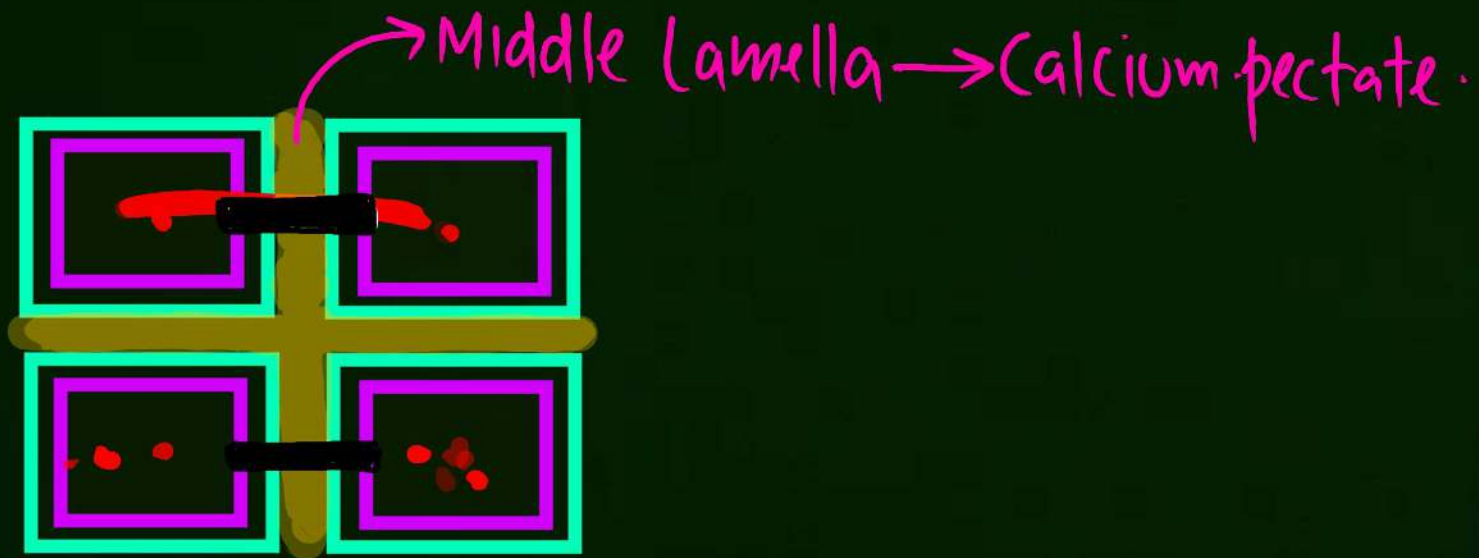
Cell wall shows-

- 1 Complete permeability.
- 2 Semi permeability
- 3 Differential permeability
- 4 Impermeability



Middle Lamella

The middle lamella is a layer mainly of **calcium pectate** which holds or glues the different neighboring cells together. The cell wall and middle lamellae may be traversed by **plasmodesmata** which connect the cytoplasm of neighboring cells.



Question: NEET-2010

Which one of the following structures between two adjacent cells is an effective transport pathway?

- 1 Plasmodesmata
- 2 Plastoquinones
- 3 Endoplasmic reticulum
- 4 Plasmalemma



Question: NEET-2009

Middle lamella is composed mainly of

- 1 Muramic acid
- 2 Calcium pectate
- 3 Phosphoglycerides
- 4 Hemicellulose



Question: NEET-2009

Plasmodesmata are-

- 1 Locomotory structures ✗
- 2 Membranes connecting the nucleus with plasmalemma ✗
- 3 Connections between adjacent cells ✓
- 4 Lignified cemented layers between cells. ✗



The Endomembrane system

Is a group of membranes and organelles in eukaryotic cells that work together to **modify, package, and transport lipids and proteins**. It's important for the cell's compartmental organization to function properly.

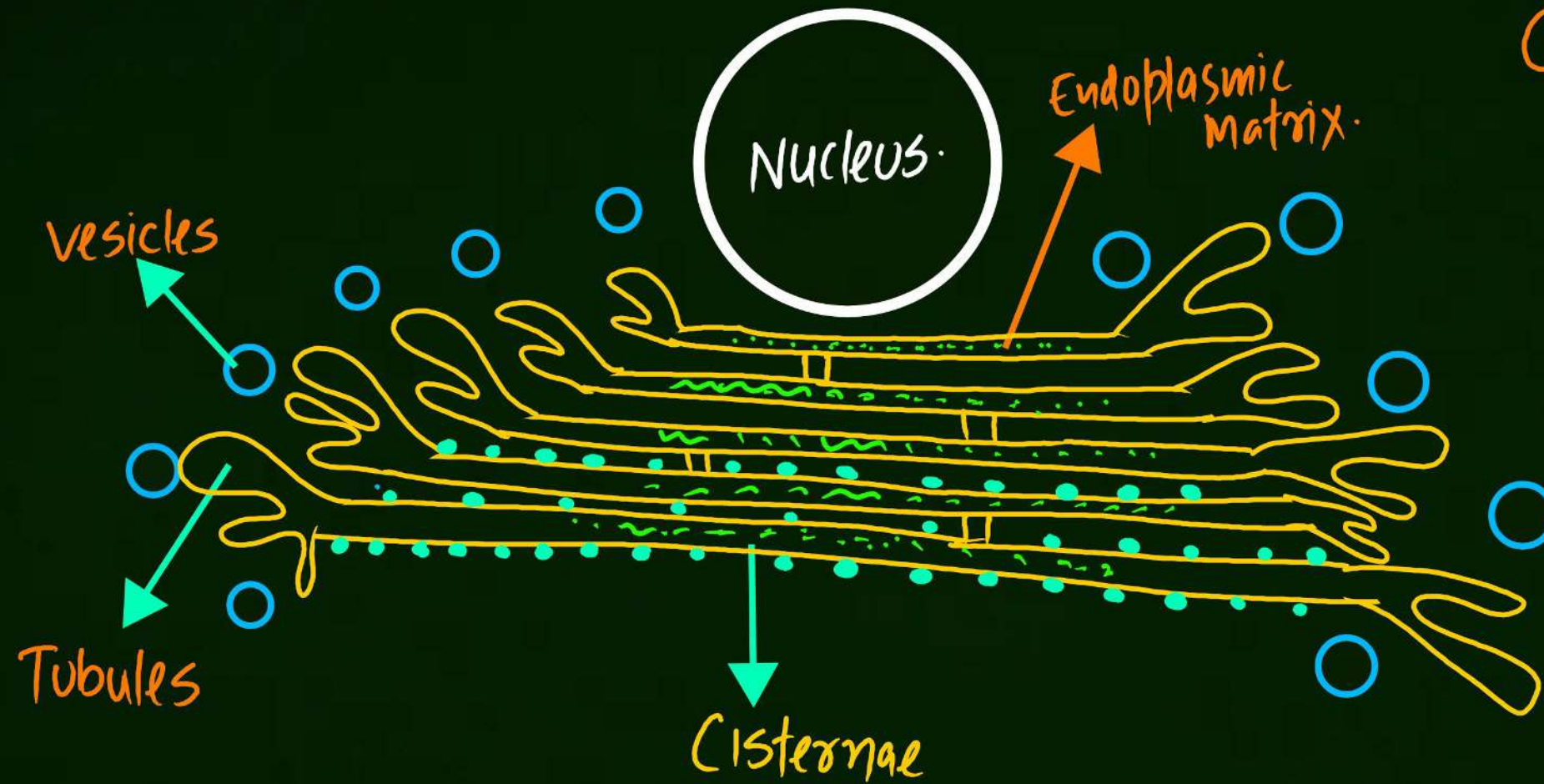
E.g. Endoplasmic Reticulum
Golgi Apparatus
Lysosome
Vacuole

Question: NEET-2021

The organelles that are included in the endomembrane system are-

- 1 Golgi complex, endoplasmic reticulum, mitochondria and lysosomes
- 2 Endoplasmic reticulum, mitochondria, ribosomes and lysosomes
- 3 Endoplasmic reticulum, Golgi complex, lysosomes and vacuoles. ✓
- 4 Golgi complex, mitochondria, ribosomes and lysosomes

The Endoplasmic Reticulum (ER)



Types

① Smooth E.R	② Rough E.R
S.E.R	R.E.R.
Aggranular E.R	Granular E.R.
↓	↓
Synthesis & Secretion of lipid	Synthesis & Secretion of protein.

The Endoplasmic Reticulum (ER)

- This is a complex network of tubes called **Cisternae**, the lumen of these Cisternae is filled with fluid called endoplasmic matrix.
- Endoplasmic reticulum has three parts: **Cisternae, Tubules & Vesicle**.
- Cisternae are long inter-connected, Flattened Sac like structures lying parallel to each other in the form of pile or stack. Inside these cisternae is a hollow space called Lumen in which is present a fluid called **Endoplasmic matrix**.
- Tubules are minute branched tube like structures attached to cisternae.
- Vesicles are bubble like structures produced at the ends of tubules and are finally pinched off to get attached with Golgi complex.

Types of Endoplasmic Reticulum (ER)

- ✓ 1. **Smooth Endoplasmic Reticulum:** (Agranular endoplasmic reticulum):- which is without ribosomes attached to its cisternae. This endoplasmic reticulum is meant to synthesize Lipids.
- ✓ 2. **Rough Endoplasmic Reticulum:** (Granular endoplasmic reticulum):- Which bears ribosomes to its surface. This endoplasmic reticulum is meant to synthesize proteins.

The functions of the endoplasmic reticulum are to form the skeletal framework of the cell, to provide a pathway for the distribution of nuclear material from one cell to the other and to synthesize fats, steroids and cholesterol with the help of enzymes secreted by the cell.

Question: NEET-2002

Which of the following statements with respect to endoplasmic reticulum is incorrect?

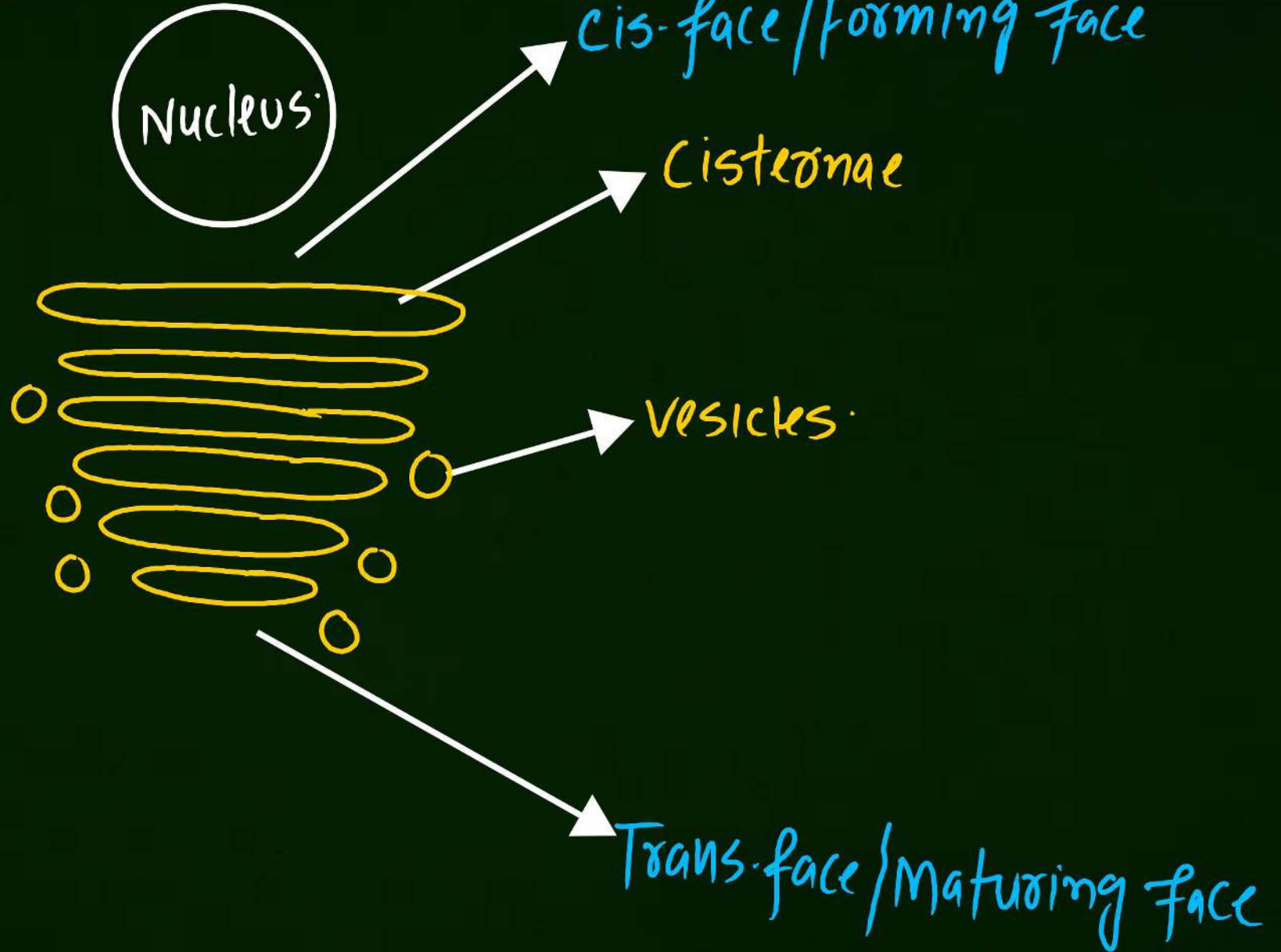
- 1 RER has ribosomes attached to ER
- 2 SER is devoid of ribosomes
- 3 In prokaryotes only RER are present.
- 4 SER are the sites for lipid synthesis



Golgi apparatus

↓
Golgi Body

→ Camello Golgi



Golgi apparatus

Camillo Golgi (1898) first observed densely stained reticular structures near the nucleus. These were later named **Golgi bodies after him**. They consist of many flat, disc-shaped sacs or cisternae and are stacked parallel to each other. Varied number of cisternae are present in a Golgi complex. The Golgi cisternae are concentrically arranged near the nucleus with distinct convex, cis or the forming face and concave, trans or the maturing face. The cis and the trans faces of the organelle are entirely different, but interconnected.

Functions of Golgi apparatus

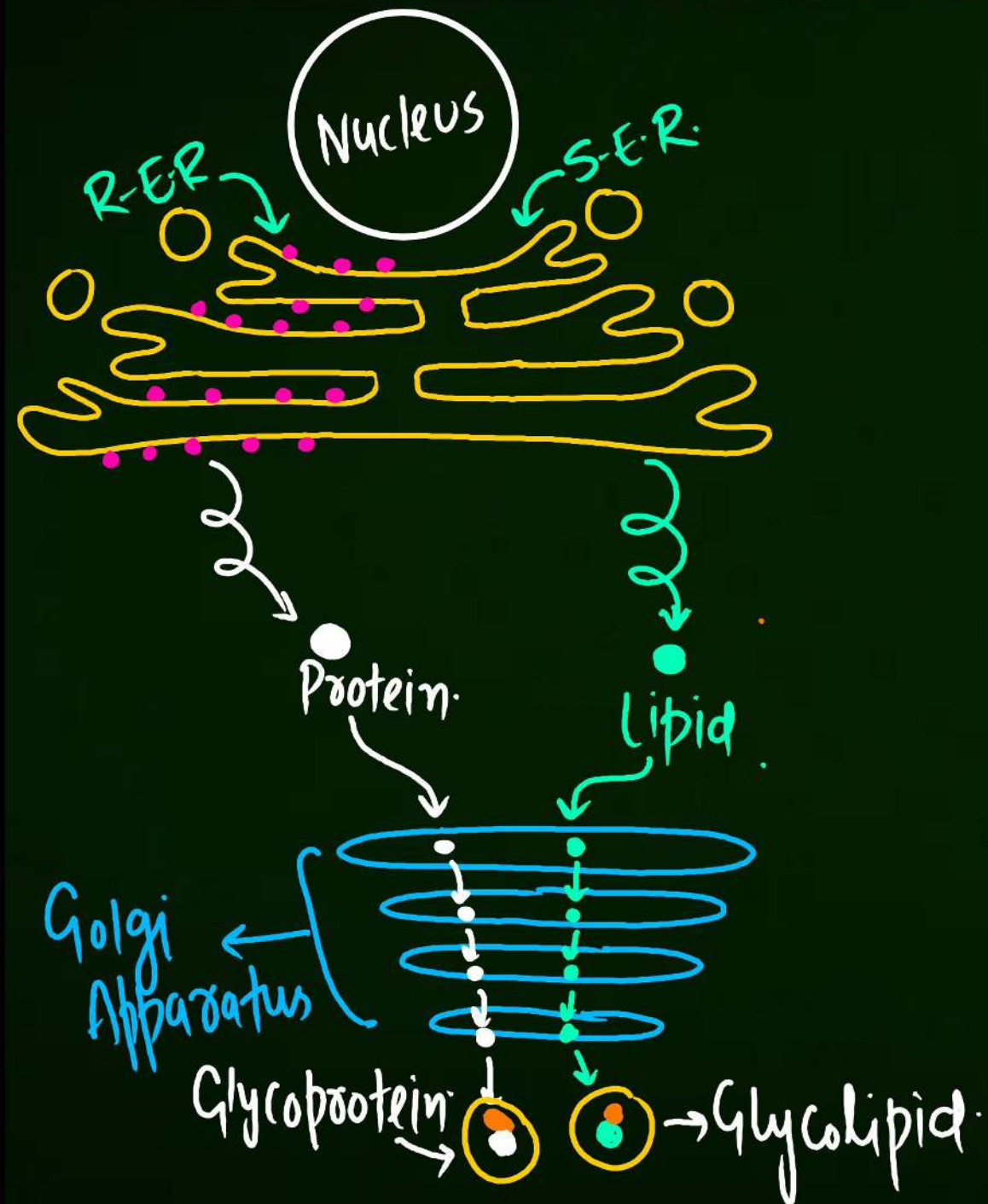
The golgi apparatus principally performs the function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell.

Materials to be packaged in the form of vesicles from the ER fuse with the cis face of the golgi apparatus and move towards the maturing face.

A number of proteins synthesised by ribosomes on the endoplasmic reticulum are **modified in the cisternae of the golgi apparatus** before they are released from its trans face.

Golgi apparatus is the important site of **formation of glycoproteins and glycolipids**.

The Endomembrane system



Question: NEET-2023

Which of the following statements are correct with respect of Golgi apparatus?

- ☒ A. It is the important site of formation of glycoprotein and glycolipids. ✓
- ☐ B. It produces cellular energy in the form of ATP. ✗
- ☒ C. It modifies the protein synthesised by ribosomes on ER. ✓
- ☐ D. It facilitates the transport of ions. ✗
- ☐ E. It provides mechanical support. ✗

Choose the most appropriate answer from the options given below.

1 B and C only

2 A and C only. ✓

3 A and D only

4 D and E only

Question: NEET-2020

8x4=32 Marks



Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells?

1 Endoplasmic reticulum

2 Peroxisomes

3 Golgi bodies

4 Polysomes



Question: NEET-2025

7x4 = 28 Marks



Assertion (A): The primary function of the Golgi apparatus is to package materials made by the endoplasmic reticulum and deliver them to intracellular targets and outside the cell. **T**

Reason (R): Vesicles containing materials made by the endoplasmic reticulum fuse with the cis face of the Golgi apparatus, and they are modified and released from the trans face of the Golgi apparatus. **T**

- 1 A is false but R is true
- ☒ 2 Both A and R are true AND R correctly explains A.
- 3 Both are true but R does not explain A
- 4 A is true but R is false

Thank
You